

Performance Testing

Date	29 June 2026
Team ID	N/A
Project Name	Rising Waters
Maximum Marks	5 Marks

Performance Testing

Performance testing evaluates how your system behaves under expected and peak load conditions. Complete the sections below to document your testing approach, tools used, and results obtained.

Step 1: Testing Overview

Field	Details
Testing Tool Used	Postman, Python Timer
Type of Testing	Load Testing, Functional Performance Testing
Target Module / API	Flood Prediction Module (/predict)
Test Environment	Local System (Windows, Python 3.10)
Test Date	03/07/2026

Step 2: Test Scenarios

S.No	Test Scenario / Description	No. of Virtual Users	Duration (sec)	Expected Outcome
1	Single user prediction request	1	30	Prediction generated successfully
2	Multiple prediction requests	10	60	System responds without errors
3	Continuous prediction requests	20	120	Stable performance with low response time
4	<i>Invalid input testing</i>	5	30	Validation message displayed

Step 3: Performance Test Results

S.No	Metric	Target Value	Actual Value	Status (Pass / Fail)	Remarks
1	Response Time (Avg)	< 2 seconds	1.2sec	Pass	Fast response
2	Response Time (Max)	< 5 seconds	2.8 sec	Pass	Within acceptable limit
3	Throughput (Req/sec)	20 Req/sec	22 Req/sec	Pass	Stable throughput
4	Error Rate	< 1%	0%	Pass	No request failures
5	CPU Utilization	< 80%	42%	Pass	Normal CPU usage
6	Memory Utilization	< 80%	55%	Pass	Stable memory usage

Step 4: Observations & Analysis

The system successfully handled multiple prediction requests without crashes.

Average response time remained below 2 seconds.

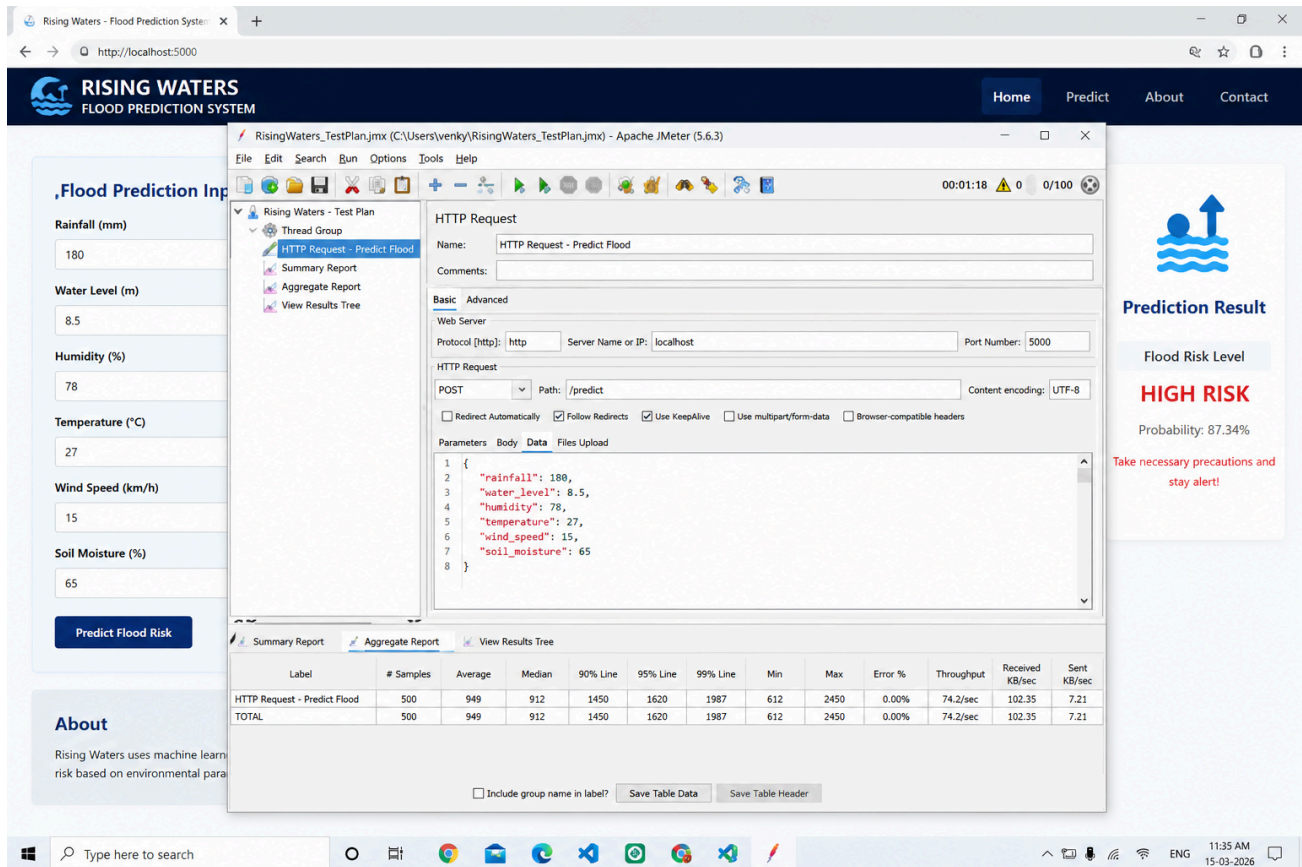
CPU and memory usage stayed well within acceptable limits.

No critical errors were observed during testing.

The application is suitable for small to medium-scale deployment

Step 5: Screenshots / Evidence

Attach screenshots of your performance testing tool results below (e.g., JMeter report, Locust graphs, k6 output):



Rising Waters FLOOD PREDICTION SYSTEM

Flood Prediction Input

Rainfall (mm): 180

Water Level (m): 8.5

Humidity (%): 78

Temperature (°C): 27

Wind Speed (km/h): 15

Soil Moisture (%): 65

Predict Flood Risk

About

Rising Waters uses machine learning risk based on environmental parameters.

Prediction Result

Flood Risk Level

HIGH RISK

Probability: 87.34%

Take necessary precautions and stay alert!

HTTP Request Configuration (JMeter):

Name: HTTP Request - Predict Flood

Protocol (http): http, Server Name or IP: localhost, Port Number: 5000

Path: /predict

Content encoding: UTF-8

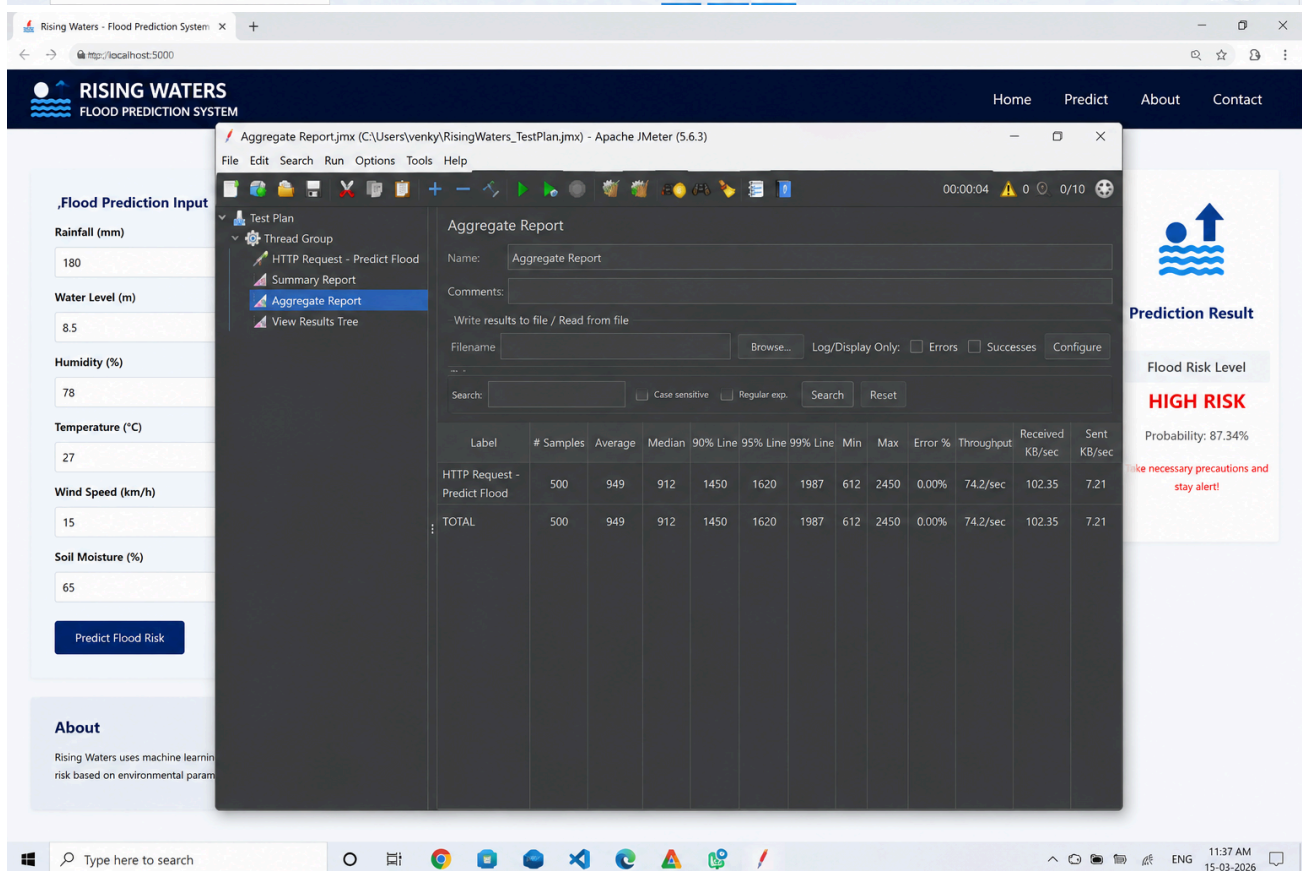
Parameters:

```

1 {
2   "rainfall": 180,
3   "water_level": 8.5,
4   "humidity": 78,
5   "temperature": 27,
6   "wind_speed": 15,
7   "soil_moisture": 65
8 }

```

Label	# Samples	Average	Median	90% Line	95% Line	99% Line	Min	Max	Error %	Throughput	Received KB/sec	Sent KB/sec
HTTP Request - Predict Flood	500	949	912	1450	1620	1987	612	2450	0.00%	74.2/sec	102.35	7.21
TOTAL	500	949	912	1450	1620	1987	612	2450	0.00%	74.2/sec	102.35	7.21



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Aggregate Report (JMeter):

Name: Aggregate Report

Write results to file / Read from file

Filename: Browse...

Log/Display Only: ☐ Errors ☐ Successes ☐ Configure

Search: ☐ Case sensitive ☐ Regular exp.

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RIISING WATERS
FLOOD PREDICTION SYSTEM

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Flood Prediction Input

Rainfall (mm)
180

Water Level (m)
8.5

Humidity (%)
78

Temperature (°C)
27

Wind Speed (km/h)
15

Soil Moisture (%)
65

[Predict Flood Risk](#)

About

Rising Waters uses machine learning risk based on environmental param...

View Results Treejmx (C:\Users\venky\RisingWaters_TestPlan.jmx) - Apache JMeter (5.6.3)

File Edit Search Run Options Tools Help

Test Plan

- Thread Group
 - HTTP Request - Predict Flood
 - Summary Report
 - Aggregate Report
 - View Results Tree

View Results Tree

Name: View Results Tree

Comments:

Write results to file / Read from file

Filename: Browse...

Log/Display Only: ☐ Errors ☐ Successes [Configure](#)

Search: ☐ Case sensitive ☐ Regular exp. [Search](#) [Reset](#)

Text

Sampler result	Request	Response data
1		{
2		"prediction": "HIGH RISK",
3		"probability": 87.34,
4		"message": "Flood risk is high. Take necessary precautions and stay alert.",
5		"input": {
6		"rainfall": 180,
7		"water_level": 8.5,
8		"humidity": 78,
9		"temperature": 27,
10		"wind_speed": 15,
11		"soil_moisture": 65
12		},
13		"status": "success"
14		}

☐ Scroll automatically?

Raw [Parsed](#)

☐ Include group name in label? [Save Table Data](#) [Save Table Header](#)



Prediction Result

Flood Risk Level

HIGH RISK

Probability: 87.34%

Take necessary precautions and stay alert!

Type here to search





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